

Iodine-125 Radiation of Posterior Uveal Melanoma

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Abstract: Twenty-eight cases of posterior choroidal melanoma were treated with iodine-125 in gold eye plaques. Eleven cases were located within 3.0 mm of the optic nerve (group A), nine were within 3.0 mm of the fovea (group B), and eight were within 3.0 mm of the optic nerve and fovea (group C). The mean follow-up of group A was 46.3 months; group B, 25.5 months; and group C, 42.7 months. Complications included macular edema, cataract and tumor growth. Visual acuity remained within two lines of that tested preoperatively for 4 of 11 patients in group A, 4 of 9 in group B, and 5 of 8 in group C. These results with iodine-125 suggest it as an appropriate treatment for patients with choroidal melanoma located near optic nerve and/or macula. [Key words: choroidal tumor, iodine-125, melanoma, radiation.] *Ophthalmology* 94:1621-1626, 1987

Iodine-125 is a low-energy emitter of gamma photons. As such, it allows one to use eye plaques made of metals of a high atomic number, such as gold, to shield normal ocular structures while radiating choroidal melanoma.¹ This is particularly important when the choroidal melanoma is located close to the optic nerve and/or macula. With other brachytherapy sources such as cobalt-60, previous authors have advised not treating the melanoma when the posterior edge of the tumor is closer than 5 mm to the optic nerve.² This article reviews the results of a small series of patients with posteriorly located choroidal melanoma. The author strongly believes that this group of patients clinically demonstrates that the radiation dose distribution to the eye is most favorable with low-energy plaques. Previous reports with cobalt-60³ and proton beam⁴ that addressed the issue of

posteriorly located tumors suffer from short-term follow-up and therefore will not be used for comparison. In addition, absence of prospective randomization precludes accurate comparison. Iodine-125 has been used at North Shore University Hospital for 7 years. We have treated 28 patients with choroidal melanoma located 3-mm or closer to the optic nerve and/or fovea with 1- to 6-year follow-up.

PATIENTS AND METHODS

The 28 patients were selected from a total of 46 of those treated at North Shore University Hospital because of their posterior location. These cases had a posterior margin of the choroidal melanoma within 3 mm of the optic nerve (11 cases, group A), 3 mm of the fovea (9 cases, group B), or 3 mm of the fovea and optic nerve (8 cases, group C). Fovea is defined as the center (foveola) of the shallow depression located in the macula. Five patients in group A had medium-sized tumors and six had large tumors. Medium is defined as having a base diameter (measured with calipers and transillumination at the time of surgery) from 10 to 15 mm or an apex height (measured by ultrasound) between 3 and 5 mm. A large melanoma had either a base diameter

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Presented at the American Academy of Ophthalmology Annual Meeting, New Orleans, November 1986.

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Table 1. Iodine-125 Cases

Case No.	Size L × W × H (mm)	Date Treated	Preoperative Visual Acuity	Postoperative Visual Acuity	Date	Interval (mos) RT to Last Visit	Complications*					Radiation Dose		Comments	Distance (mm) to Disc/Fovea	
							Cataract	Macular Edema	Tumor Growth	Vitreous Hemorrhage	Died of Metastases	Died of Other Causes	Dose to Apex (cGy)			Dose Rate (cGy)
Group A† (mean follow-up, 46.3 mos)																
1	9 × 7 × 4.0	2/78	20/20	CF	5/80	27	—	—	Yes	—	7/80	—	9200	77	—	4.5/3.0
6	11 × 6 × 3.0	7/78	CF	NLP	—		—	—	Yes	—	—	6/82	8760	73	Enucleated 2/80	1.5/0.0
7	8 × 7 × 7.0	10/78	20/40	20/60	10/84	72	34	29	—	—	—	—	10000	42	RD, spontaneous resolution	1.5/6.5
9	12 × 14 × 6.0	6/79	20/200	LP	11/84	65	12	—	—	—	—	1/85	11000	72	—	0.0/1.0
12	7 × 6 × 3.0	8/79	20/20	20/20	5/85	69	—	—	—	—	—	—	11800	160	—	0.0/3.5
15	16 × 16 × 10.0	4/80	20/40	NLP	5/81		—	—	—	—	—	—	8000	37	Neovascular glaucoma, enucleation 5/81	0.0/3.0
19	8 × 8 × 5.0	11/80	20/25	20/400	11/85	60	—	—	—	—	—	—	9600	60	—	3.0/6.5
23	12 × 11 × 7.0	10/81	20/200	20/400	7/85	45	—	—	Yes	#	—	—	10000	50	—	1.5/5.0
25	12 × 12 × 5.0	12/81	20/40	NLP	7/85	43	11	—	—	1	—	—	9000	65	RD	3.0/7.0
38	11 × 11 × 7.0	8/83	20/40	20/30	7/85	23	—	—	—	—	—	—	8000	80	—	5.0/1.5
40	8 × 8 × 3.0	4/84	20/100	HM	5/85	13	—	—	—	5	—	—	10000	68	—	4.0/0.5
Group B† (mean follow-up, 25.5 mos)																
5	18 × 16 × 6.0	6/78	20/50	20/100	6/79	12	—	—	—	—	2/80	—	8150	54	—	4.5/2.0
26	10 × 10 × 7.0	1/82	20/70	NLP	1/84		19	—	Yes	—	—	—	8000	50	Enucleated 1/84	5.0/1.0
28	9 × 9 × 5.0	6/82	20/40	20/50	7/85	37	—	—	—	—	—	—	8000	45	—	4.5/1.0
30	11 × 11 × 3.5	7/82	20/40	20/200	7/85	36	—	13	—	—	—	—	12600	100	—	9.0/1.0
31	15 × 15 × 6.0	8/82	20/50	LP	7/85	35	19	—	—	1	—	—	10080	60	—	3.5/1.0
33	12 × 12 × 6.0	8/82	20/20	20/200	8/85	36	14	7	—	—	—	—	9600	80	—	4.0/0.0
39	10 × 10 × 5.0	9/83	20/50	20/70	9/85	24	—	—	—	—	—	—	9720	45	—	1.0/4.5
43	12 × 12 × 7.0	5/84	20/25	20/25	5/85	12	—	—	—	—	—	—	8000	80	—	4.5/1.0
45	12 × 12 × 6.5	5/84	20/100	20/100	5/85	12	—	—	—	—	—	—	9240	55	—	5.0/1.0
Group C§ (mean follow-up, 42.7 mos)																
8	11 × 11 × 6.0	6/79	LP	LP	5/85	71	—	—	—	—	—	—	7000	60	—	3.0/0.0
13	20 × 16 × 7.0	9/79	20/100	20/70	2/81	17	—	—	—	—	9/81	—	6400	38	—	1.0/1.5
20	9 × 9 × 4.0	11/80	20/30	CF	4/85	54	23	—	—	1	—	—	8400	50	—	0.0/1.0
21	16 × 14 × 6.0	4/81	HM	CF	5/85	49	—	—	—	—	—	—	7400	112	—	3.0/1.5
24	5 × 8 × 8.0	10/81	20/70	CF	5/85	43	—	—	—	—	—	—	10000	60	—	2.0/0.0
29	12 × 12 × 5.0	6/82	20/70	20/70	5/85	35	—	—	—	—	—	—	9600	100	—	2.0/1.0
36	14 × 10 × 6.0	1/83	20/200	20/200	7/85	30	—	23	—	—	—	—	11000	90	—	0.5/0.0
37	12 × 12 × 4.0	6/83	20/20	NLP	9/84		—	—	Yes	—	—	—	9600	100	Enucleated 9/84	0.5/1.0

RT = radiation therapy; CF = counting fingers; LP = light perception; HM = hand motions; NLP = no light perception; RD = retinal detachment.

* Onset noted in months from treatment.

† Within 3 mm of disc but >3 mm from fovea.

‡ Within 3 mm of fovea but >3 mm from disc.

§ Within 3 mm of disc and fovea.

|| Enucleated.

Retreated with iodine-125 and microwave hyperthermia.

greater than 15 mm or height greater than 5 mm. Of the patients in group B, three had medium and six had large choroidal melanomas. Three patients in group C had medium and five had large choroidal melanomas. Base diameter has been the best prognostic indication of mortality, therefore, it is worth noting that only one case in both groups A and B and two cases in group C had a base diameter greater than 15 mm. All patients had a minimum follow-up of 1 year. Table 1 details all cases. The location was documented with fundus photography and fluorescein angiography. We used 1 disc diameters (DD) as being equivalent to 1.5 mm.

All patients had preoperative evaluation for metastatic disease (liver profile, computed tomography of liver, brain, and orbit, and chest x-ray). Preoperative determination of tumor base diameters was based on fluorescein angiograms, and height of tumor was measured with A-scan ultrasonography. These measurements were necessary to determine the size of the gold plaque. Consultation with a radiation therapist preceded surgical placement of the iodine-125/gold plaque. The preference was for plaques 4 to 6 mm larger than the estimated base diameter.⁵ A total dose of 6400 to 12,600 cGy was delivered to the tumor apex; the dose rate was between 37 and 160 cGy per hour (1 rad = 1 cGy). Twenty-one of 28 cases fell between 8,000 and 10,000 rad. The technique has been described previously and was similar to that used for other radioactive plaques. The placement of the plaque was performed with the patient under general anesthesia and was removed using local anesthesia. A 90° to 180° peritomy was performed in the area of the tumor. Transillumination was used to locate the tumor margin which was marked on the sclera with gentian violet. The measurement was always within 1 mm of the preoperative size estimate made with fluorescein angiograms. The edges where the perimeter of the plaque and suture holes were to be located were also marked on the sclera. Mersilene sutures (5-0) secured the plaque to the sclera. Conjunctiva was closed with one running 6-0 plain catgut suture. No dummy plaques were used, and the P-32 uptake test was not performed.

I examined all patients after surgery. The interval for follow-up examinations varied depending on the status of the eye. All patients were examined at least annually. Repeated examinations included some or all of the following: visual acuity, visual fields, biomicroscopy, ocular tensions, gonioscopy, ophthalmoscopy (direct, indirect, and fundus contact), ultrasonography (A- and B-scan), and fluorescein angiography. Macular edema was defined with fluorescein angiography.

The complications that were due to radiation included cataract, macular edema, tumor growth, and vitreous hemorrhage. Dilated examinations were performed at least twice during the first year at 6 and 12 months after irradiation. Results of the preoperative examination disclosed the presence of any cataract (cortical or posterior subcapsular). After the first year, there

were yearly dilated examinations specifically for cataract.

Macular edema and tumor growth were identified with ophthalmoscopy and fluorescein angiography which were performed annually. Vitreous hemorrhage that decreased the patient's vision was recorded often at the time of occurrence rather than at the annual examination. Vitreous hemorrhage that did not cause loss of vision was noted at the time of routine annual evaluation. The cause of death was determined for all cases by phone discussions with the patient's physician at the time of death.

RESULTS

Of the 28 cases, 11 were within 3.0 mm of the optic nerve (group A), 9 were within 3.0 mm of the fovea (group B), and 8 were within 3.0 mm of the optic nerve and fovea (group C). The mean follow-up of group A was 46.3 months; group B, 25.5 months, and group C, 42.7 months.

Tumor size and total radiation dose and dose rate are found in the table. Macular edema occurred at 7, 13, 23, and 29 months after treatment in cases 33, 30, 29, and 7, respectively. Cataract occurred in seven cases at a mean of 19 months after irradiation. Cataract was identified when first noted, not when it caused a decrease in vision. Angiographic macular edema occurred in four cases (7, 29, 30, 33). A cataract developed in three cases in group A, in three in group B, and in one in group C. Tumor growth was noted in five cases; three in group A, one in group B, and one in group C.

Visual acuity remained within two lines of the preirradiation vision in 4 of 11 cases from group A, 4 of 9 from group B, and 5 of 8 in group C. At the 1-year follow-up, 75% of all cases (28) had vision within two lines of pretreatment vision. At 2 years follow-up, 14 of 22 cases (64%) had retained vision within two lines of pretreatment (3 years, 10 of 16 cases [63%]; 4 years, 6 of 11 cases [55%]; and 5 years, 3 of 7 cases [43%]). Cases 6 and 15 from group A were enucleated, Case 26 from group B was enucleated, and case 37 from group C was enucleated. One case from each group died of melanoma metastasis at intervals from treatment of 20, 24, and 29 months. Thus, 3 of 28 cases died of metastatic melanoma for 89% survival with a mean follow-up of 39 months. The sizes of the melanoma in these three cases were $9 \times 7 \times 4$, $18 \times 16 \times 6$, and $20 \times 16 \times 7$ mm. The only other significant complication was vitreous hemorrhage which occurred in three cases at 1, 1, and 5 months after treatment. This does not necessarily represent a complication of radiation but rather the surgical procedure or the tumor's own propensity to hemorrhage.

Figure 1 represents the visual acuity in four cases before and after treatment. This demonstrates several

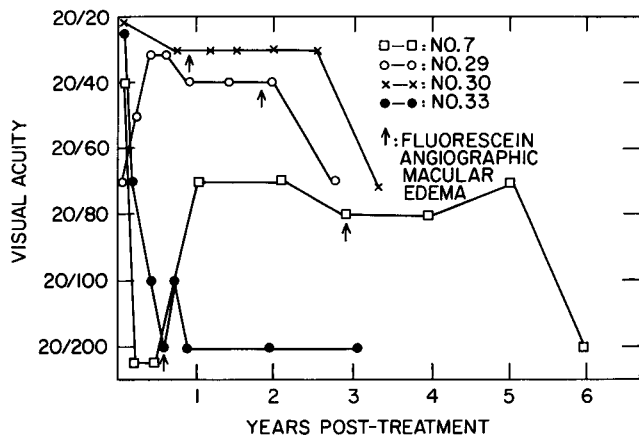


Fig 1. The courses of cases 7, 29, 30, and 33 can be seen over a period of time.

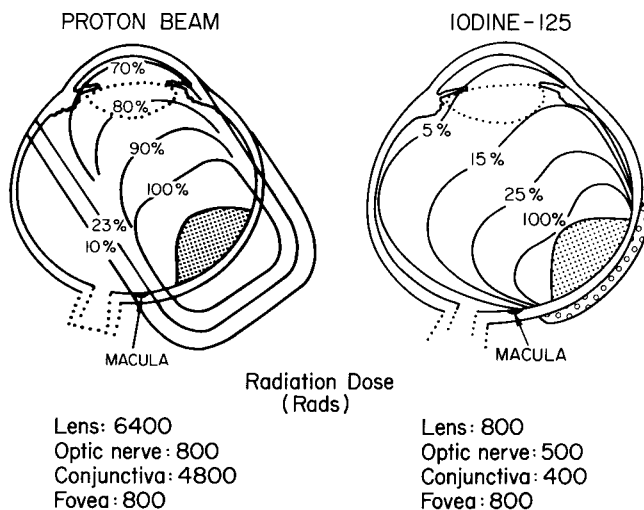


Fig 2. A comparison of the radiation dose to various ocular structures when treating a posterior uveal melanoma with proton beam (left)⁷ and iodine-125 (right).

points. Case 7 shows that vision may decline immediately after treatment from vitreous hemorrhage and then improve. This case also illustrates that the angiographic macular edema seen 3 years after treatment did not cause a decrease in vision; and when vision decreased, it was difficult to be certain whether it was due to macular edema or the cataract that had developed.

The improvement in vision in case 29 was due to the decrease in macular distortion that had been caused by the tumor. The macular edema noted in case 30 did coincide with the decrease in vision. Finally, the decrease in vision in case 33 was due to macular edema and cataract formation which developed 7 and 14 months, respectively, after irradiation.

DISCUSSION

Previous reports of radiation treatment of posteriorly located choroidal melanoma with cobalt-60^{2,3} and proton beam⁴ cannot be used for comparison with these data for many reasons. The data were not collected prospectively or randomly and therefore are subject to bias. In addition, the risk factors other than size would need to be included (e.g., exact location of anterior and posterior tumor margin). The latter information is not available and therefore precludes meaningful comparison.

One is therefore left with the basic principles of radiation dose distribution which would indicate rather clearly that the anterior segment of the eye receives considerably more radiation when treating posteriorly located choroidal melanoma with proton beam than with iodine-125. Recent analysis⁶ of patients treated with proton beam yields slightly modified results from previous reports⁷ but no substantial change as far as radiation to anterior structures of the eye.

Figure 2 represents a tumor less than 3 mm from the fovea for both proton beam and iodine-125. The more recent report for proton beam⁶ gives dosimetry for a tumor more than 3 mm (2.5 DD) from the optic nerve.

A valid comparison of iodine-125 to cobalt-60 is similarly thwarted; but again the physical principles of radiation would discourage one from using cobalt-60 plaques for posteriorly located choroidal melanoma.² It is crucial in understanding this problem to realize that any radioactive isotope is able to deliver sterilizing doses of radiation to any size choroidal melanoma. The advantages of using low-energy isotopes such as iodine-125 are that (1) they allow shielding of vital ocular structures adjacent to the tumor, such as the optic nerve, with a thin metal eye plaque (this attenuation can be appreciated by the fact that it takes 5 mm of platinum to reduce the radiation from cobalt-60 by 50%, whereas <0.01 mm reduces the radiation from iodine-125 by 50%), and (2) the inverse square law indicates, and our laboratory data confirm, that in a simulated experiment where both cobalt-60 and iodine-125 deliver the same dose to the apex of a 5-mm tumor, the normal ocular structures 15 to 20 mm away from the plaques receive 30% more radiation with cobalt-60.⁸

Improvements in iodine-125/gold plaque design will allow an even better distribution of radiation as far as normal ocular structures are concerned. A collimator has been designed to reduce the penumbra (stray radiation). Recessing the iodine-125 seeds allows for further collimation of the radiation. Both of these maneuvers allow equalizing the tumor apex-to-base dose. The latter is only a potential problem with large tumors but a more uniform dose distribution within the tumor can certainly be achieved with iodine-125/gold scleral plaques.

It must be recognized that radiation complications become apparent with time. Reporting patients with less than 1 year of follow-up is misleading. This is supported

by our noting that at 1 year after treatment of 28 cases, 75% had retained vision within two lines of that noted before radiation therapy. The percentages of cases retaining vision (within two lines of pretreatment) fell to 43% at 5 years. I am optimistic concerning the relative merit of iodine-125 versus other forms of radiation therapy (cobalt-60,³ ruthenium-106⁹ or proton beam^{4,10}) but would caution that with long-term follow-up (5–10 years) the complication rate will rise and be substantial. Although cataracts are usually removed with a high degree of success, the long-term results in irradiated eyes are not known. One could imagine less than optimal outcome in eyes with poor wound healing—a possibility, or deficient ocular surface lubrication. Radiation maculopathy, optic atrophy, and neovascularization will cause permanent loss of vision. Optic atrophy has not been seen in any patients treated with iodine-125. Radiation maculopathy, neovascularization, and cataract will occur in an almost predictable fashion, depending on tumor size, location, and length of follow-up.

It is clear that only a prospective randomized trial would yield an accurate comparison of the various radiation sources. However, this report supports the use of iodine-125 for posterior uveal melanoma. Among its many advantages are ready availability and the use of

techniques that are easily learned by most ophthalmic surgeons and radiation therapists.

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Discussion

by

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Radiation sources for brachytherapy have included radon-222 seeds, gold-198 seeds, cobalt-60, iridium-192 seeds, ruthenium-106, and iodine-125 seeds. Although most of the American experience in brachytherapy for uveal melanoma has been with cobalt-60, in recent years the isotope iodine-125 has been considered by some investigators to be a more desirable radiation source because of its advantageous physical characteristics which result in the singular advantage that its radiation can be almost totally shielded with a thin plate of gold approximately 0.5-mm thick. We took advantage of this shielding capacity by designing the iodine-125 therapeutic plaque with a thin gold backing having a shape somewhat similar to that of a bottle cap. Since the radiation from iodine-125 can be essentially totally shielded, the potential radiation hazard to persons handling the plaque is minimized, and a preferential directional radiation pattern can be concentrated toward the tumor in direct line with the unshielded iodine-125 while at the same time shielding tissues behind the plaque. Both because of the real and the theoretical advantages of iodine-125, proponents of its use have been hopeful that the incidence of radiation retinopathy and optic neuropathy will be lessened when com-

pared with other radiation sources or at least that the onset of these complications will be delayed.

In Dr. Packer's retrospective review of 28 cases with posteriorly located melanomas that were treated with iodine-125 and followed for a mean interval of approximately 3 years, he has attempted to identify the incidence of radiation complications that affect visual acuity through the development of macular edema, optic neuropathy, cataract, and other complications including enucleation. It is noteworthy that almost half of the patients retained central vision within two lines of the preoperative visual acuity measurements. Unfortunately, Dr. Packer does not identify which of the eyes losing vision had radioactive plaques placed directly behind the fovea. Since therapeutic iodine-125 plaques are designed to be 4 to 6 mm larger in diameter than the base diameter of the tumor, it is conceivable that some of the eyes with tumors located within 2 to 3 mm from the fovea will receive direct foveal exposure from a plaque whose edge is located immediately behind the fovea. Other information not included in Dr. Packer's report is the presence or absence of serous detachment of the sensory retina at the macula. Meaningful comparisons of complications between radiation sources are virtually impossible unless comparisons are made between eyes harboring tumors of similar size and location. In addition, it is difficult to compare results when the complications are reported differently.

In future studies, I believe it will be worthwhile for ophthalmologists and radiotherapists to work in concert to calculate as accurately as possible the theoretical radiation exposure to the tumor apex, optic nerve, fovea, and the crystalline lens. Documentation should include the size of the tumor, its location, and specifically the distance from its posterior perimeter to the fovea and the optic nerve and the distance of the center of the tumor from the fovea and optic nerve. We plan to profile these data in the National Collaborative Ocular Melanoma Study. Eventually, we hope information will become available that will provide us with a better understanding of the relationship between observed complications and the theoretical calculations of dose to the vital structures listed above.

I congratulate Dr. Packer for being among the first in this country to use the isotope iodine-125 in the management of choroidal melanoma and for reporting in this review, not only

the effectiveness of iodine-125 in promoting tumor shrinkage but the specific complications that relate to loss of vision in this series of cases followed for an average of approximately 3 years. Dr. Packer concludes that the incidence of visual complications observed after iodine-125 use in the treatment of melanomas located within 3 mm of the optic nerve or fovea compares favorably with complications observed after other radiation sources including cobalt-60. However, the numbers of cases reported were small and the method of reporting complications was not strictly comparable to reports with other isotopes. In the future, it will be increasingly important that clinicians who are using other isotopes in brachytherapy, as well as those using charged particles, attempt to report their complications in a uniform manner so that real differences in complications will become evident with the accumulation of carefully collected data in years to come.